Generalized Variational Inference (GVI)

Posterior beliefs with the rule of three

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Structure of the talk

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   - 1.2 The optimization perspective
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1 The form of the Bayesian problem

Purpose of part 1: Motivate the rule of three

(1) Bayesian inference minimizes **losses**

(2) Bayesian inference **regularizes** with the prior

(3) Bayesian inference = **optimization** over **(sub)spaces of probability measures**

1.1 The Bayesian problem: Traditional perspective

Ingredients (for the simplest case) are:

- $n = n_1 + n_2$ observations $\boldsymbol{x} = (x_1, x_2, \ldots, x_{n_1+n_2})^T$,
- **prior** $\pi(\boldsymbol{\theta})$,
- **likelihoods** $\{p(x_i|\boldsymbol{\theta})\}_{i=1}^{n_1+n_2}$

Output = **posterior belief**:

$$q^*(\boldsymbol{\theta}) \propto \pi(\boldsymbol{\theta}) \prod_{i=1}^{n_1+n_2} p(x_i|\boldsymbol{\theta}) = \widetilde{\pi}(\boldsymbol{\theta}) \prod_{i=n_1+1}^{n_2} p(x_i|\boldsymbol{\theta}), \quad \text{for } \widetilde{\pi}(\boldsymbol{\theta}) = \pi(\boldsymbol{\theta}) \prod_{i=1}^{n_1} p(x_i|\boldsymbol{\theta})$$

Inference interpretation = belief updates:

- likelihoods $\{p(x_i|\boldsymbol{\theta})\}_{i=1}^{n_1+n_2}$ update prior about $\boldsymbol{\theta}$
- Old posterior $\widetilde{\pi}(\boldsymbol{\theta})$ = new prior (**coherence**/**Bayesian additivity**)

1.2 The Bayesian problem: The optimization perspective

Zellner (1988) shows that the Bayes posterior $q^*(\boldsymbol{\theta})$ solves

$$q^*(\boldsymbol{\theta}) = \underset{q \in \mathcal{P}(\boldsymbol{\Theta})}{\arg\min} \left\{ \underbrace{\mathbb{E}_{q(\boldsymbol{\theta})} \left[\sum_{i=1}^{n} -\log(p(x_i|\boldsymbol{\theta})) \right]}_{\text{minimized by } q(\boldsymbol{\theta}) = \delta_{\hat{\boldsymbol{\theta}}_n}(\boldsymbol{\theta}),\ \hat{\boldsymbol{\theta}}_n = \text{MLE}} + \underbrace{\mathrm{KLD}\,(q||\pi)}_{\text{minimized by } q = \pi} \right\}, \quad (1)$$

Notation:

- $\mathcal{P}(\boldsymbol{\Theta})$ = all probability distributions on $\boldsymbol{\Theta}$
- KLD = Kullback-Leibler divergence = $\mathbb{E}_{q(\boldsymbol{\theta})} [\log q(\boldsymbol{\theta}) - \log \pi(\boldsymbol{\theta})]$

Inference interpretation = regularized loss-minimization:

- $-\log(p(x_i|\boldsymbol{\theta}))$ = **loss** of $\boldsymbol{\theta}$ for x_i
- Inference = regularizing MLE $\hat{\boldsymbol{\theta}}_n$ with $\mathrm{KLD}(q||\pi)$

1.3 The Bayesian problem: The loss-minimization perspective

Bissiri et al. (2016): Bayes posteriors $q^*(\boldsymbol{\theta})$ for general loss $\ell(\boldsymbol{\theta}, x_i)$:

$$q^*(\boldsymbol{\theta}) \propto \pi(\boldsymbol{\theta}) \exp\left\{-\sum_{i=1}^{n_1+n_2} \ell(\boldsymbol{\theta}, x_i)\right\} = \widetilde{\pi}(\boldsymbol{\theta}) \exp\left\{-\sum_{i=n_1+1}^{n_2} \ell(\boldsymbol{\theta}, x_i)\right\}$$

$$\text{for } \widetilde{\pi}(\boldsymbol{\theta}) = \pi(\boldsymbol{\theta}) \exp\left\{-\sum_{i=1}^{n_1} \ell(\boldsymbol{\theta}, x_i)\right\}$$

Inference interpretation = belief updates:

- Again: losses $\{\ell(\boldsymbol{\theta}, x_i)\}_{i=1}^{n_1+n_2}$ update prior about $\boldsymbol{\theta}$
- Again: Old posterior $\widetilde{\pi}(\boldsymbol{\theta})$ = new prior (**coherence**)
- Difference: $\boldsymbol{\theta}$ arbitrary, e.g. $\ell(\boldsymbol{\theta}, x_i) = |x_i - \boldsymbol{\theta}|$ admissible

1.4 The Bayesian problem: The new perspective I/II

Easy to show: Zellner's representation valid for any $\ell(\boldsymbol{\theta}, x_i)$:

$$q^*(\boldsymbol{\theta}) = \underset{q \in \mathcal{P}(\boldsymbol{\Theta})}{\arg\min} \left\{ \underbrace{\mathbb{E}_{q(\boldsymbol{\theta})} \left[\sum_{i=1}^{n} \ell(\boldsymbol{\theta}, x_i) \right]}_{\text{minimized by } \delta_{\hat{\boldsymbol{\theta}}_n}(\boldsymbol{\theta})} + \underbrace{\mathrm{KLD}\left(q || \pi\right)}_{\text{minimized by } q = \pi} \right\}$$

Bissiri et al. (2016)'s generalization (preserves coherence):

- Replacing $-\log(p(x_i|\boldsymbol{\theta}))$ with other losses $\ell(\boldsymbol{\theta}, x_i)$

Two more generalizations (break coherence):

- Replacing $\mathcal{P}(\boldsymbol{\Theta})$ with $\mathcal{Q} \subset \mathcal{P}(\boldsymbol{\Theta})$ ($=$ **VI**)
- Replacing KLD with inference-problem specific regularizers

1.4 The Bayesian problem: The new perspective II/II

Our **generalized representation of Bayesian inference**:

$$q^*(\boldsymbol{\theta}) = \arg\min_{q \in \Pi} \left\{ \underbrace{\mathbb{E}_{q(\boldsymbol{\theta})}\left[\sum_{i=1}^{n} \ell(\boldsymbol{\theta}, x_i)\right]}_{\text{minimized by } \delta_{\hat{\boldsymbol{\theta}}_n}(\boldsymbol{\theta})} + \underbrace{D\left(q \| \pi\right)}_{\text{minimized by } q = \pi} \right\}$$

Notation:

- if $\boldsymbol{\Pi}$ = variational family, write $\mathcal{Q}$.
- $\ell_n(\boldsymbol{\theta}, \boldsymbol{x}) = \sum_{i=1}^{n} \ell(\boldsymbol{\theta}, x_i)$

Inference interpretation = regularized & constrained minimization:

- $\boldsymbol{\ell}_n(\boldsymbol{\theta}, \boldsymbol{x})$ = **loss** of $\boldsymbol{\theta}$ to minimize
- **D** = **divergence**, acting as uncertainty quantifier/regularizer
- $\boldsymbol{\Pi}$ = set of **admissible posterior** beliefs
- **Inference** = constrained, regularized optimization

  $\Rightarrow$ **Shorthand Notation:** $P(\boldsymbol{\ell}_n, D, \boldsymbol{\Pi})$

2 The form of the Generalized Bayesian problem

Purpose of part 2: Investigate $P(\boldsymbol{\ell}_n, D, \boldsymbol{\Pi})$

(1) Interpretations & modularity of $\boldsymbol{\ell}_n$, D and $\boldsymbol{\Pi}$?

(2) Is there an axiomatic justification?

(3) Which existing methods does this (not) encompass?

2.1 Generalized Bayesian problem: provable modularity

$$q^*(\boldsymbol{\theta}) = \arg\min_{q \in \Pi} \left\{ \underbrace{\mathbb{E}_{q(\boldsymbol{\theta})} \left[\sum_{i=1}^{n} \ell(\boldsymbol{\theta}, x_i) \right]}_{\text{minimized by } \delta_{\hat{\boldsymbol{\theta}}_n}(\boldsymbol{\theta})} + \underbrace{D\left(q || \pi\right)}_{\text{minimized by } q = \pi} \right\}$$

Roles of ℓ_n, D, Π::

- ℓ_n: which parameter $\boldsymbol{\theta}$ do we care about?
- **D**: How is uncertainty quantified/what does q^* look like?
- Π: Which beliefs are allowed?

  $\Rightarrow$ (provable) modularity of $P(\ell_n, D, \Pi)$!

Theorem 1 (GVI modularity)

For Bayesian inference with $P(\ell_n, D, \Pi)$, making it robust to model misspecification amounts to changing ℓ_n. Conversely, adapting uncertainty quantification (fixing Π, π, $\boldsymbol{\theta}^*$, $\hat{\boldsymbol{\theta}}_n$) amounts to changing D.

2.2 Generalized Bayesian problem: Axiomatic derivation I/II

Axiom 1 (Representation)

Bayesian inference infers posteriors q on $\boldsymbol{\Theta}$ by (i) measuring how q fits a sample $\boldsymbol{x}$ via the expectation of a loss $\ell_n(\boldsymbol{\theta}, \boldsymbol{x})$, (ii) quantifying uncertainty about $\boldsymbol{\theta}^*$ via a divergence D between prior π and q, (iii) optimizing q over a space of probability distributions Π on $\boldsymbol{\Theta}$.

Axiom 2 (Information Difference)

$P(\ell_n, D, \Pi)$ produces different posteriors for $\boldsymbol{x} = x_{1:n}$ and $\boldsymbol{x}' = x_{1:n+m}$ if there is an information difference, i.e. if $\ell_n(\theta, \boldsymbol{x}) \neq \ell_{n+m}(\theta, \boldsymbol{x}')$.

Axiom 3 (Prior Regularization)

q is regularized against π by penalizing the divergence $D(q||\pi)$.

Axiom 4 (Translation Invariance)

For constant C and $\ell'_n = \ell_n + C$, $P(\ell'_n, D, \Pi) = P(\ell_n, D, \Pi)$.

2.2 Generalized Bayesian problem: Axiomatic derivation II/II

Theorem 2 (Form 1)

If Axiom 1 holds, $P(\ell_n, D, \Pi)$ has form $\arg\min_{q \in \Pi} \{L(q|\boldsymbol{x}, \ell_n, D)\}$ for $L(q|\boldsymbol{x}, \ell_n, D) = f\left(\mathbb{E}_{q(\boldsymbol{\theta})}[\ell_n(\boldsymbol{\theta}, \boldsymbol{x})], D(q||\pi)\right)$, for some $f : \mathbb{R}^2 \to \mathbb{R}$.

Theorem 3 (Form 2)

For $P(\ell_n, D, \Pi)$ being $\arg\min_{q \in \Pi} \{L(q|\boldsymbol{x}, \ell_n, D)\}$ and $\circ$ an elementary operation on $\mathbb{R}$, $L(q|\boldsymbol{x}, \ell_n, D) = \mathbb{E}_{q(\boldsymbol{\theta})}[\ell_n(\boldsymbol{\theta}, \boldsymbol{x})] \circ D(q||\pi)$ satisfies Axioms 3 and 4 only if $\circ = +$.

Implications/relevance:

- Bayesian inference = constrained, regularized optimization
- Objective only depends on $\mathbb{E}_{q(\boldsymbol{\theta})}[\ell_n(\boldsymbol{\theta}, \boldsymbol{x})]$ and $D(q||\pi)$
- For elementary $f\left(\mathbb{E}_{q(\boldsymbol{\theta})}[\ell_n(\boldsymbol{\theta}, \boldsymbol{x})], D(q||\pi)\right)$, f must be addition.

  (**Note:** Axiom 4 excludes most non-elementary f)

2.3 Generalized Bayesian problem & existing methods I/III

$$q^*(\boldsymbol{\theta}) = \arg\min_{q \in \boldsymbol{\Pi}} \left\{ \mathbb{E}_{q(\boldsymbol{\theta})} \left[\ell_n(\boldsymbol{\theta}, \boldsymbol{x}) \right] + D\left(q || \pi\right) \right\}$$

$P(\ell_n, D, \boldsymbol{\Pi})$ covers & gives **insight** into existing methods, e.g.

- **Power Bayes**: $P(w\ell_n, D, \Pi) = P(\ell_n, \frac{1}{w} D, \Pi)$.
  ($\Longrightarrow$ w-power likelihood $= \frac{1}{w} \times$ **more trust in your prior**.)
- **Regularized Bayes**: Adding $\Phi(q(\boldsymbol{\theta}, \boldsymbol{x})) = \mathbb{E}_{q(\boldsymbol{\theta}, \boldsymbol{x})} \left[\phi(\boldsymbol{\theta}, \boldsymbol{x}) \right]$ into the objective corresponds to $P\left(\ell_n + \phi, D, \Pi\right)$.
  ($\Longrightarrow$ RegBayes = a form of **GVI** that changes ℓ_n)

2.3 Generalized Bayesian problem & existing methods II/III

Method	$\ell(\boldsymbol{\theta}, x_i)$	D	Π
Standard Bayes	$-\log(p(\boldsymbol{\theta}\|x_i))$	KLD	$\mathcal{P}(\boldsymbol{\Theta})$
Generalized Bayes[1]	any ℓ	KLD	$\mathcal{P}(\boldsymbol{\Theta})$
Power Bayes[2]	$-\log(p(\boldsymbol{\theta}\|x_i))$	$\frac{1}{w}\mathrm{KLD},\ w > 1$	$\mathcal{P}(\boldsymbol{\Theta})$
Divergence Bayes[3]	divergence-based ℓ	KLD	$\mathcal{P}(\boldsymbol{\Theta})$
Standard VI	$\boldsymbol{-\log\left(p(\theta\|x_i)\right)}$	**KLD**	$\mathcal{Q}$
Power VI[4]	$-\log(p(\boldsymbol{\theta}\|x_i))$	$\frac{1}{w}\mathrm{KLD},\ w > 1$	$\mathcal{Q}$
Regularized Bayes[5]	$-\log(p(\boldsymbol{\theta}\|x_i)) + \phi(\boldsymbol{\theta}, x_i)$	KLD	$\mathcal{Q}$
Gibbs VI[6]	any ℓ	KLD	$\mathcal{Q}$
Generalized VI	**any ℓ**	**any D**	$\mathcal{Q}$

Table 1 – $P(\ell_n, D, Q)$ & existing methods. [1](Bissiri et al., 2016), [2](e.g. Holmes and Walker, 2017; Grünwald et al., 2017; Miller and Dunson, 2018), [3](e.g. Hooker and Vidyashankar, 2014; Ghosh and Basu, 2016; Futami et al., 2017; Jewson et al., 2018), [4](e.g. Yang et al., 2017; Huang et al., 2018) [5](Ganchev et al., 2010; Zhu et al., 2014), [6](Alquier et al., 2016; Futami et al., 2017)

2.3 Generalized Bayesian problem & existing methods III/III

Not everything fits $P(\ell_n, D, \mathbf{\Pi})$:

(1) **Laplace approximations** (e.g., INLA)

(2) **F-Variational inference (F-VI)**: VI based on discrepancy $F \neq \mathrm{KLD}$ (locally) solving $q^* = \arg\min_{q \in \mathcal{Q}} F(q \| \widetilde{q})$ for $\widetilde{q}$ = standard Bayesian posterior, e.g.

F = Rényi's α-divergence (Li and Turner, 2016; Saha et al., 2017)
F = χ-divergence (Dieng et al., 2017)
F = operators (Ranganath et al., 2016)
F = scaled AB-divergence (Regli and Silva, 2018)
F = Wasserstein distance (Ambrogioni et al., 2018)
...

(3) **Expectation Propagation (EP)** (Minka, 2001; Opper and Winther, 2000) and its variants (e.g. Hernández-Lobato et al., 2016).
Note: Particular type of **F-VI**, with F = (local) reverse KLD

3 Reinterpreting VI

Purpose of part 3: Motivating **GVI**

(1) **Standard VI**: Optimality & reinterpretation

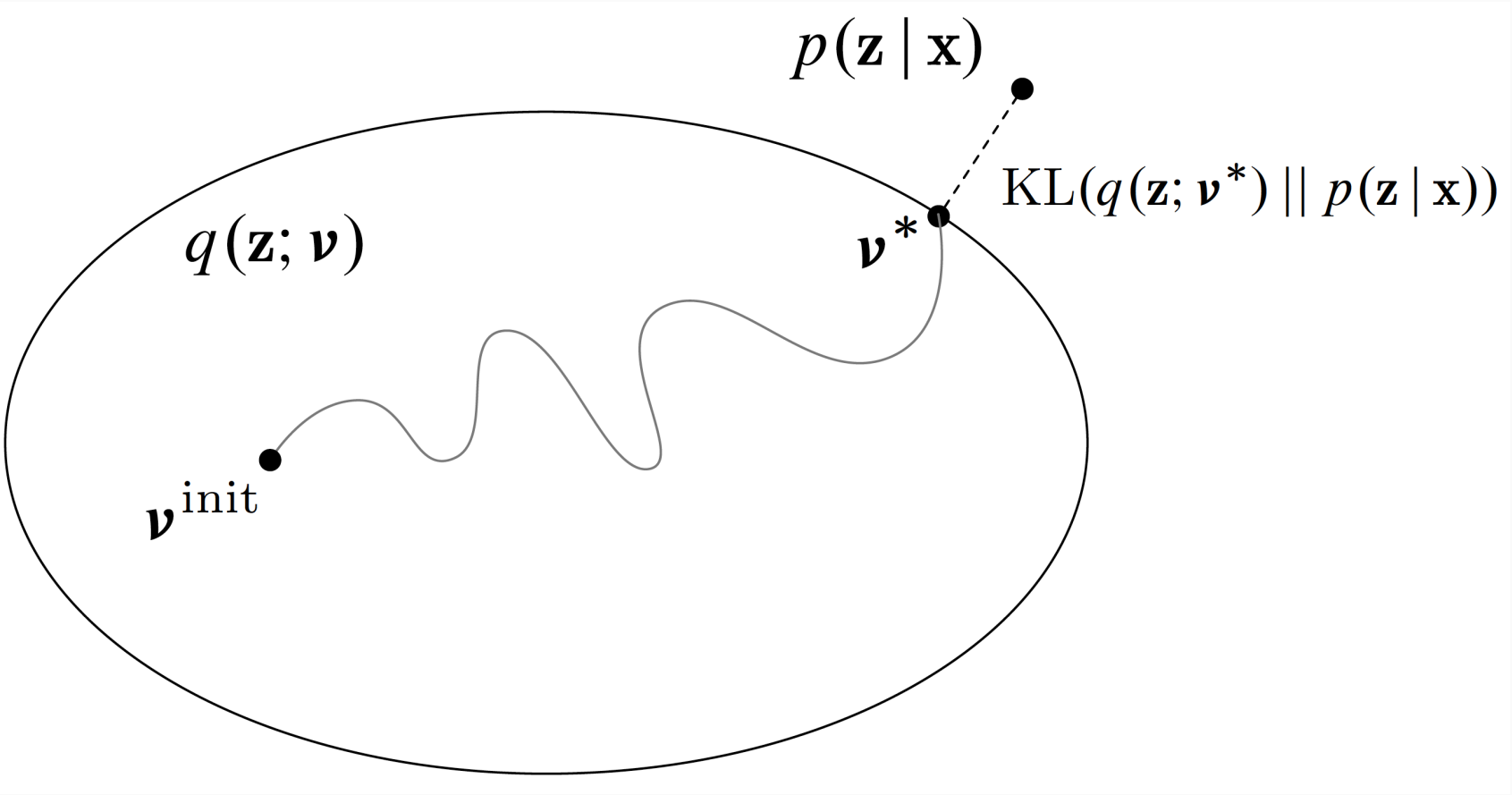
(2) **F-VI**: "suboptimal" methods with better posteriors

(3) **GVI**: A modular alternative to **F-VI**

3.1 Optimality & reinterpretation of standard VI I/V

Relationship between VI and exact inference?

Traditional view: Discrepancy-minimization, i.e. **VI** = approximation minimizing the KLD to $\widetilde{q}$. (Inspiration for **F-VI** methods)



[From *Variational Inference: Foundations and Innovations* (Blei, 2019)]

3.1 Optimality & reinterpretation of standard VI II/V

Standard VI: $q^* = \arg\min_{q \in \mathcal{Q}} \mathrm{KLD}(q \| \widetilde{q})$, $\widetilde{q}$ solves $P(\ell_n, \mathbf{KLD}, \mathcal{P}(\Theta))$

$$\mathrm{KLD}(q || \widetilde{q}) =$$

$$\underbrace{\mathbb{E}_{q(\boldsymbol{\theta})}\left[\log\left(\frac{q(\boldsymbol{\theta})}{\exp\left\{-\sum_{i=1}^n \ell(\boldsymbol{\theta}, x_i)\right\}\pi(\boldsymbol{\theta})}\right)\right]}_{\text{(Generalized) ELBO}} + \underbrace{\log\left(\int_{\boldsymbol{\theta}} \exp\left\{-\sum_{i=1}^n \ell(\boldsymbol{\theta}, x_i)\right\}\pi(\boldsymbol{\theta}) d\boldsymbol{\theta}\right)}_{\text{Generalized 'log evidence'}}$$

Inference = minimizing ELBO, which you can rewrite as

$$\mathrm{ELBO}(q) = \mathbb{E}_{q(\boldsymbol{\theta})}\left[\sum_{i=1}^n \ell(\boldsymbol{\theta}, x_i)\right] + \mathrm{KLD}(q || \pi). \tag{2}$$

. . . which is exactly the objective of $P(\ell_n, \mathbf{KLD}, \mathcal{Q})$

3.1 Optimality & reinterpretation of standard VI III/V

In other words, $P(\ell_n, \mathbf{KLD}, \mathcal{Q})$ $(= \text{ELBO})$ is

$$q^*(\boldsymbol{\theta}) = \underset{q \in \mathcal{Q}}{\arg\min} \left\{ \mathbb{E}_{q(\boldsymbol{\theta})} \left[\ell_n(\boldsymbol{\theta}, \boldsymbol{x}) \right] + \text{KLD}\left(q || \pi\right) \right\},$$

the $\mathcal{Q}$-constrained relaxation of $P(\ell_n, \mathbf{KLD}, \mathcal{P}(\Theta))$, whose objective is

$$q^*(\boldsymbol{\theta}) = \underset{q \in \mathcal{P}(\Theta)}{\arg\min} \left\{ \mathbb{E}_{q(\boldsymbol{\theta})} \left[\ell_n(\boldsymbol{\theta}, \boldsymbol{x}) \right] + \text{KLD}\left(q || \pi\right) \right\},$$

(which is the exact Bayesian objective).

$\Rightarrow$ Reinterpretation of **standard VI** as **Constrained optimization**!

3.1 Optimality & reinterpretation of standard VI IV/V

Alternative view: VI $=$ $\mathcal{Q}$-constrained version of exact Bayes problem

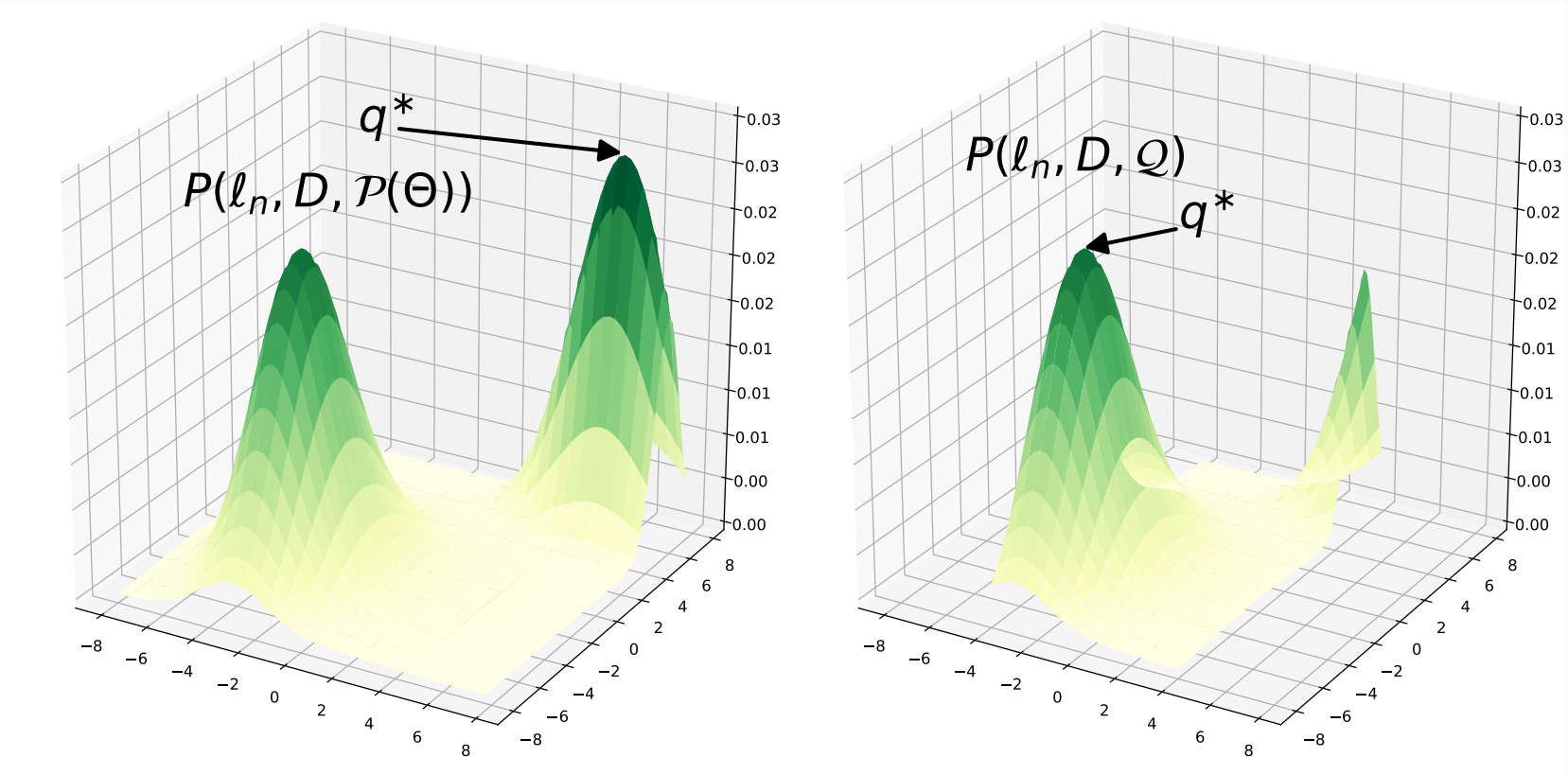


Figure 1 – Left: Unconstrained (i.e. exact) Bayesian inference. **Right:** Constrained (i.e. standard variational) Bayesian inference

3.1 Optimality & reinterpretation of standard VI V/V

Consequence I/II: VI-optimality

Theorem 4 (VI optimality)

For exact and coherent Bayesian posteriors solving $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\boldsymbol{\Theta}))$ and a fixed variational family $\mathcal{Q}$, standard VI produces the uniquely optimal $\mathcal{Q}$-constrained approximation to $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\boldsymbol{\Theta}))$ Having decided on approximating the Bayesian posterior with some $q \in \mathcal{Q}$, VI provides the uniquely optimal solution.

3.2 Why does F-VI produce better posteriors? I/II

Consequences II/II: F-VI-suboptimality. **Three** big disadvantages:

(1) If $F \neq \mathrm{KLD}$, F-VI violates Axioms 1–4.

(2) F-VI conflates ℓ_n and D (i.e., modularity of $P(\ell_n, D, \boldsymbol{\Pi})$ lost).

(3) Last Thm: F-VI gives worse $\mathcal{Q}$-constrained posterior than standard VI (*relative to the standard Bayesian problem* $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\boldsymbol{\Theta}))$)

Objection! F-VI often produces better posteriors than **standard VI**!

3.2 Why does F-VI produce better posteriors? II/II

Seeming contradiction:

(1) VI is the best approximation to the *standard* Bayesian posterior

(2) F-VI often outperforms VI (e.g., on test scores)

Resolution:

F-VI outperforms VI by implicitly **targeting a non-standard Bayesian problem** that is more appropriate than $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\Theta))$

$\Longrightarrow$ Inspires **Generalized Variational Inference (GVI)**

3.3 Towards GVI I/II

GVI = combining advantages of **VI** and **F-VI**:

(1) Has form $P(\ell_n, D, \mathcal{Q})$ Like **VI** i.e.
   (i) satisfies Axioms 1–4;
   (ii) provably interpretable modularity (**loss, uncertainty quantifier, admissible posteriors**

(2) Derives different & more appropriate posteriors like **F-VI** but
   (i) without conflating ℓ_n and D
   (ii) with **explicit** rather than **implicit** changes .

Definition 1 (GVI)

Any Bayesian inference method solving$P(\ell_n, D, \mathcal{Q})$ with admissible choices ℓ_n, D and $\mathcal{Q}$ is a Generalized Variational Inference (**GVI**) method satisfying Axioms 1 – 4.

3.3 Towards GVI II/II

Illustration: F-VI aims for D, but changes ℓ_n – GVI doesn't

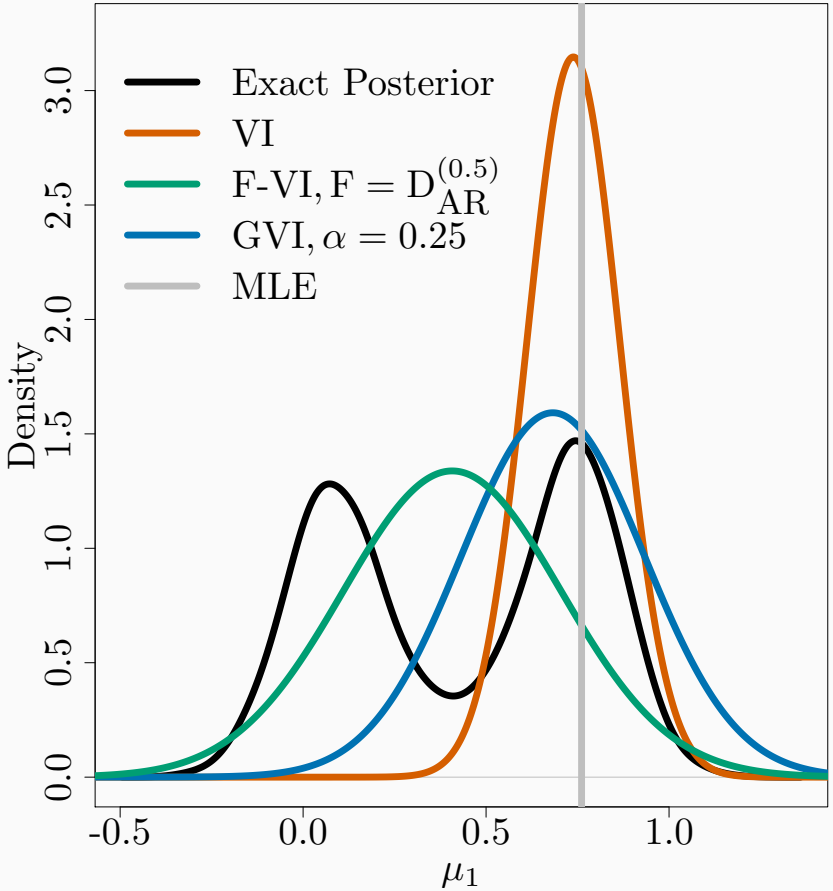


Figure 2 – **Exact**, VI, F-VI ($F = D_{AR}^{(0.5)}$) and $P(\ell_n, D_{AR}^{(\alpha)}, \mathcal{Q})$ based GVI marginals of the location in a 2 component mixture model. Respecting ℓ_n, VI and GVI provide uncertainty quantification around the most likely value $\widehat{\boldsymbol{\theta}}_n$ via D. In contrast, F-VI implicitly changes the loss and has a mode at the locally most *unlikely* value of $\boldsymbol{\theta}$.

4 GVI: What does it do?

Purpose of part 4: Exploring three use cases of **GVI**

(1) Robust alternatives to $\ell(\boldsymbol{\theta}, x_i) = -\log(p(x_i|\boldsymbol{\theta}))$

(2) Prior-robust uncertainty quantification via D

(3) Adjusting marginal variances via D

4.1 GVI: The losses I/III

GVI modularity: The loss ℓ_n

Q1: Why use $\ell_n(\boldsymbol{\theta}, \boldsymbol{x}) = \sum_{i=1}^n -\log(p(x_i|\boldsymbol{\theta}))$?
A: Assuming that the true data-generating mechanism is $\boldsymbol{x} \sim g$,

$$\arg\min_{\boldsymbol{\theta}} \sum_{i=1}^n -\log(p(x_i|\boldsymbol{\theta})) \approx \arg\min_{\boldsymbol{\theta}} \mathbb{E}_g \left[-\log(p(\boldsymbol{x}|\boldsymbol{\theta}))\right]$$
$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_g \left[-\log\left(p(\boldsymbol{x}|\boldsymbol{\theta})\right) + \log(g(\boldsymbol{x}))\right] = \arg\min_{\boldsymbol{\theta}} \mathrm{KLD}(g \| p(\cdot|\boldsymbol{\theta}))$$

Interpretation: $\boldsymbol{-\log\left(p(x_i|\theta)\right)}$ = targeting KLD-minimizing $p(\cdot|\boldsymbol{\theta})$

Q2: Are there other $\boldsymbol{\mathcal{L}^D(p(x_i|\theta))}$ for divergence $\boldsymbol{D}$?
A: Yes! (e.g. Jewson et al., 2018; Futami et al., 2017; Ghosh and Basu, 2016; Hooker and Vidyashankar, 2014)

4.1 GVI: The losses II/III

Q3: Why use other $\boldsymbol{\mathcal{L}^D(p(x_i|\theta))}$?
A: Robustness (for $\boldsymbol{D}$ = a robust divergence) [log/KLD non-robust!]

Robustness recipe: $\alpha/\beta/\gamma$-divergences using generalized log functions
E.g.: β indexes β-divergence ($D_B^{(\beta)}$) via

$$\log_\beta(x) = \frac{1}{(\beta-1)\beta}\left[\beta x^{\beta-1} - (\beta-1)x^\beta\right]$$
$$D_B^{(\beta)}(g||p(\cdot|\boldsymbol{\theta})) = \mathbb{E}_g\left[\log_\beta(p(\boldsymbol{x}|\boldsymbol{\theta})) - \log_\beta(g(\boldsymbol{x}))\right]$$

Note 1: $D_B^{(\beta)} \to \mathrm{KLD}$ as $\beta \to 1$!
Note 2: Admits $D_B^{(\beta)}$-targeting loss as

$$\boldsymbol{\mathcal{L}_p^\beta(\theta, x_i)} = -\frac{1}{\beta-1}p(x_i|\boldsymbol{\theta})^{\beta-1} + \frac{I_{p,\beta}(\boldsymbol{\theta})}{\beta}, \quad I_{p,c}(\boldsymbol{\theta}) = \int p(x|\boldsymbol{\theta})^c dx$$

4.1 GVI: The losses III/III

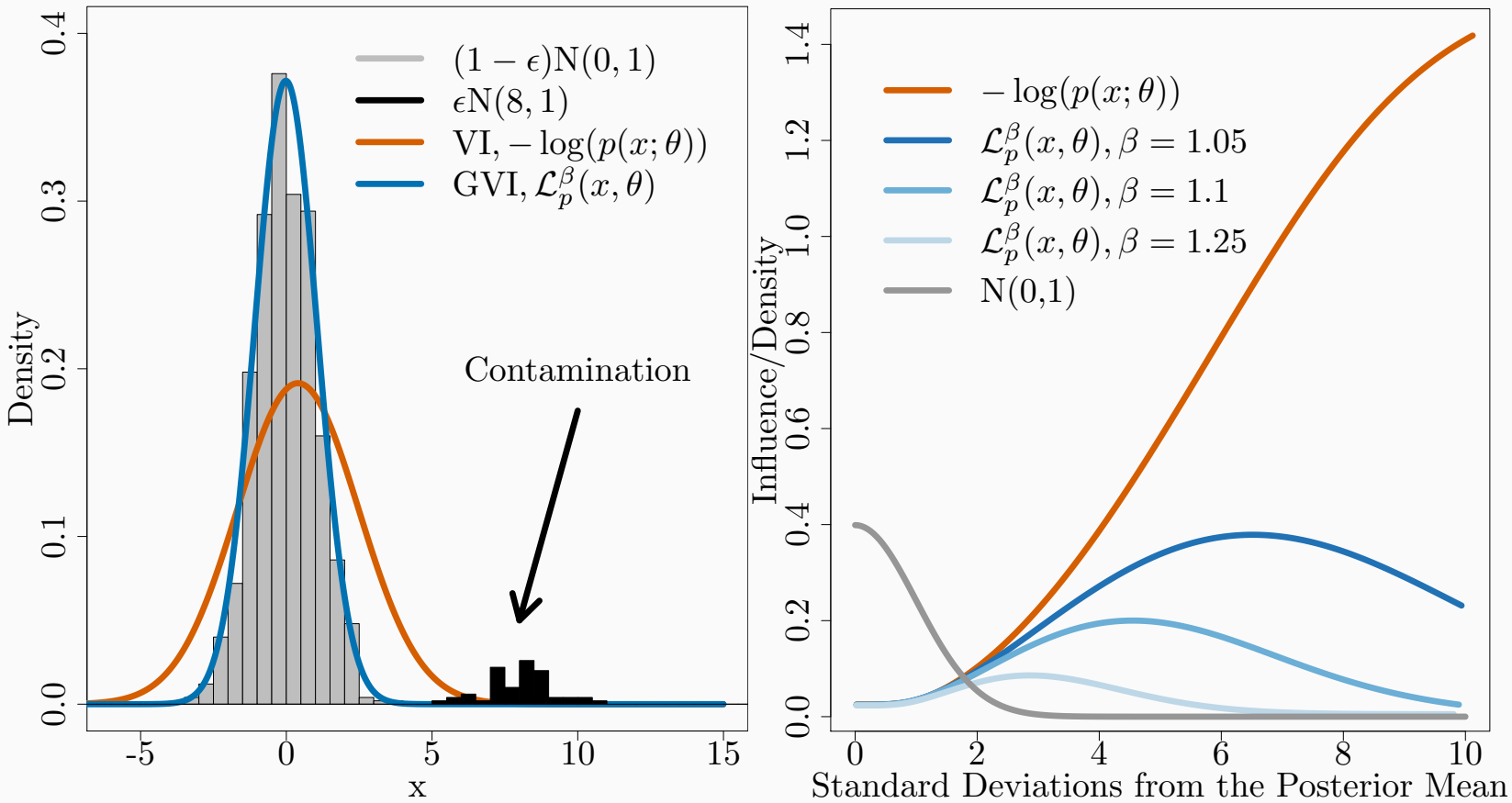


Figure 3 – Left: Robustness against model misspecification. Depicted are posterior predictives under $\varepsilon = 5\%$ outlier contamination using **VI** and $P(\sum_{i=1}^n \mathcal{L}_p^\beta(\boldsymbol{\theta}, x_i), \mathrm{KLD}, \mathcal{Q})$, $\beta = 1.5$. **Right:** From Knoblauch et al. (2018). Influence of x_i on exact posteriors for different losses.

4.2 GVI: Uncertainty Quantification I/III

GVI modularity: The uncertainty quantifier D

Q: Which VI drawbacks can be addressed via D?

A: Any **uncertainty quantification** properties, e.g.

- Over-concentration (= underestimating marginal variances)
- Sensitivity to badly specified priors
- ...

4.2 GVI: Uncertainty Quantification II/III

Example 1: GVI can fix over-concentrated posteriors

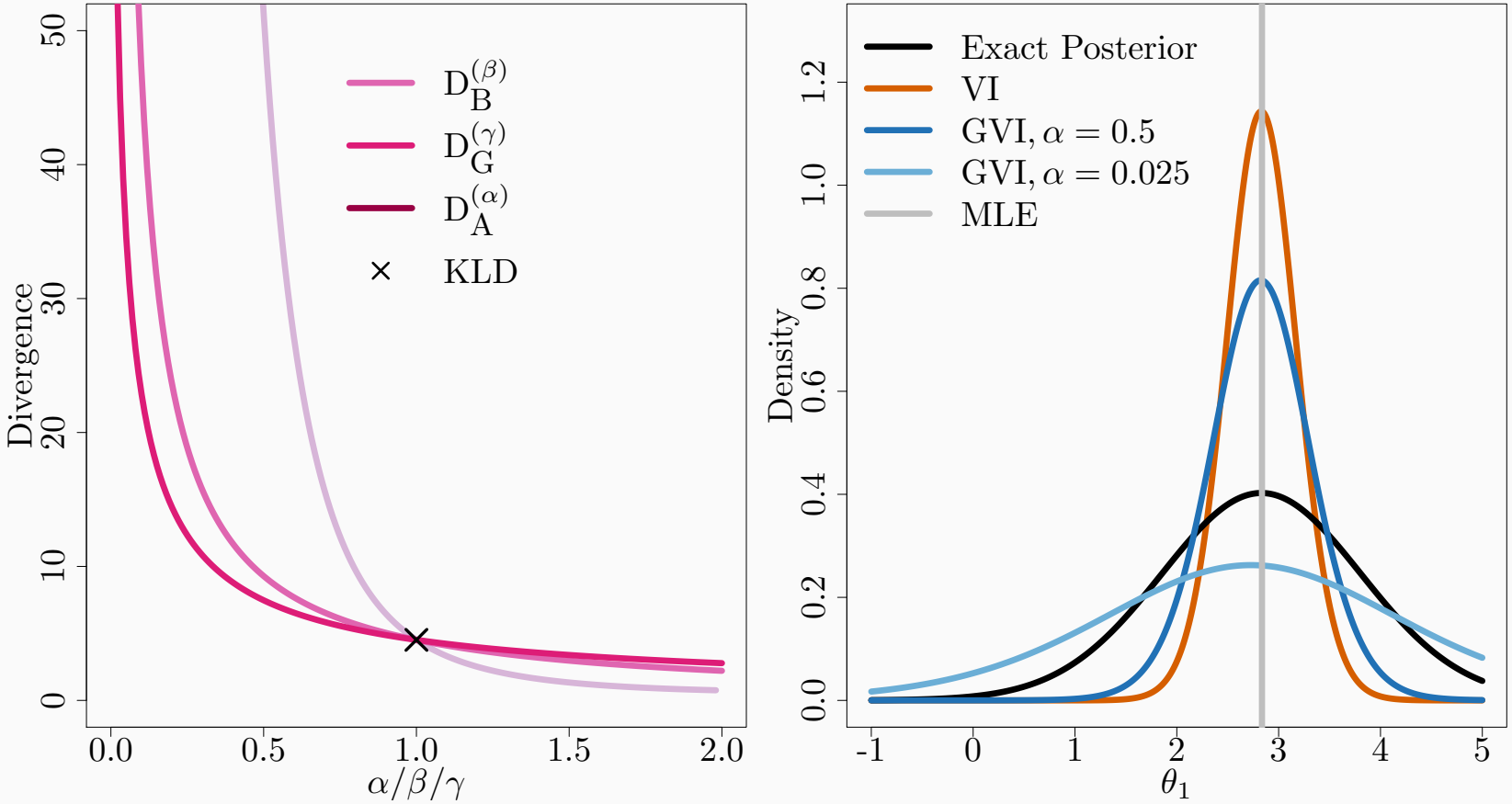


Figure 4 – Left: Magnitude of the penalty incurred by $D(q||\pi)$ for different uncertainty quantifiers D and fixed densities π, q. **Right**: Using $D_{AR}^{(\alpha)}$ with different choices of α to "customize" uncertainty.

4.2 GVI: Uncertainty Quantification III/III

Example 2: Avoiding prior sensitivity

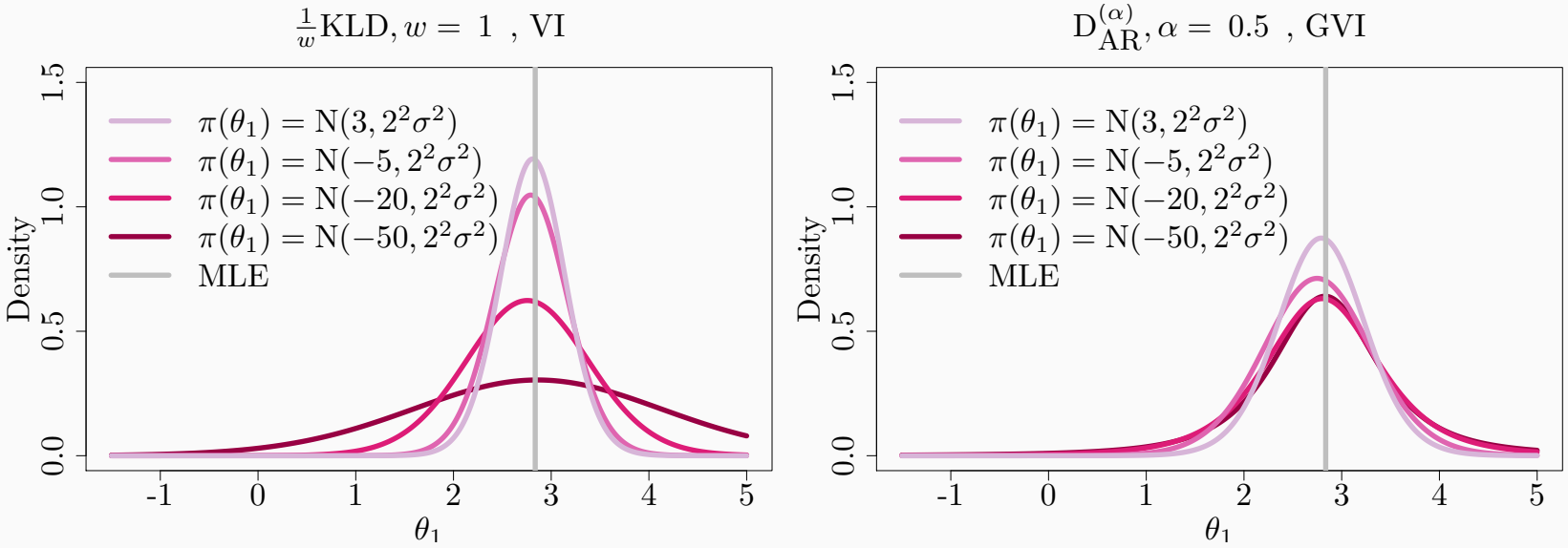


Figure 5 – Prior sensitivity with **VI** (**left**) vs. prior robustness with **GVI** (**right**). Priors are more badly specified for **darker** shades.

4.3 GVI: Three use cases

Summary: some **GVI** applications include

(1) Robustness to model misspecification ($=$ adapting ℓ_n)

(2) "Customized" marginal variances ($=$ adapting D)

(3) Prior robustness ($=$ adapting D)

5 GVI: Inference & Experiments

Purpose of part 5: **GVI** inference & experiments

(1) How/when can we "black box" **GVI**?

(2) **F-VI** vs **GVI** & changes in D (on Bayesian Neural Nets)

(3) **VI** vs **GVI** & changes in ℓ_n (on Deep Gaussian Processes)

5.1 Black Box GVI

Setup: $\mathcal{Q} = \{q(\boldsymbol{\theta}|\boldsymbol{\kappa}) : \boldsymbol{\kappa} \in K\}$ variational family s.t.

(i) one can sample $\boldsymbol{\theta}^{(1:S)} \sim q(\boldsymbol{\theta}|\boldsymbol{\kappa})$;

(ii) derivative $\nabla_{\boldsymbol{\kappa}} \log(q(\boldsymbol{\theta}|\boldsymbol{\kappa}))$ exists.

Case 1: Closed form for $\nabla_{\boldsymbol{\kappa}} D(q||\pi)$ $\rightarrow$ unbiased estimate:

$$\nabla_{\boldsymbol{\kappa}} \hat{L}(q|\ell_n, D) = \frac{1}{S} \sum_{s=1}^{S} \left\{ \ell_n(\boldsymbol{\theta}^{(s)}, \boldsymbol{x}) \cdot \nabla_{\boldsymbol{\kappa}} \log(q(\boldsymbol{\theta}^{(s)}|\boldsymbol{\kappa})) \right\} + \nabla_{\boldsymbol{\kappa}} D(q||\pi)$$

Thm. 7: Closed forms for most $\alpha/\beta/\gamma$- & Rényi-divergence.

Case 2: $D(q||\pi) = \mathbb{E}_{\boldsymbol{q}}[\ell^{D}_{\boldsymbol{\kappa},\pi}(\boldsymbol{\theta})]$ (e.g., f-divs) $\rightarrow$ unbiased estimate:

$$\nabla_{\boldsymbol{\kappa}} \hat{L}(q|\ell_n, D) = \frac{1}{S} \sum_{s=1}^{S} \left\{ \left[\ell_n(\boldsymbol{\theta}^{(s)}, \boldsymbol{x}) + \ell^{D}_{\boldsymbol{\kappa},\pi}(\boldsymbol{\theta}^{(s)}) \right] \cdot \nabla_{\boldsymbol{\kappa}} \log(q(\boldsymbol{\theta}^{(s)}|\boldsymbol{\kappa})) \right.$$
$$\left. + \nabla_{\boldsymbol{\kappa}} \ell^{D}_{\boldsymbol{\kappa},\pi}(\boldsymbol{\theta}^{(s)}) \right\}.$$

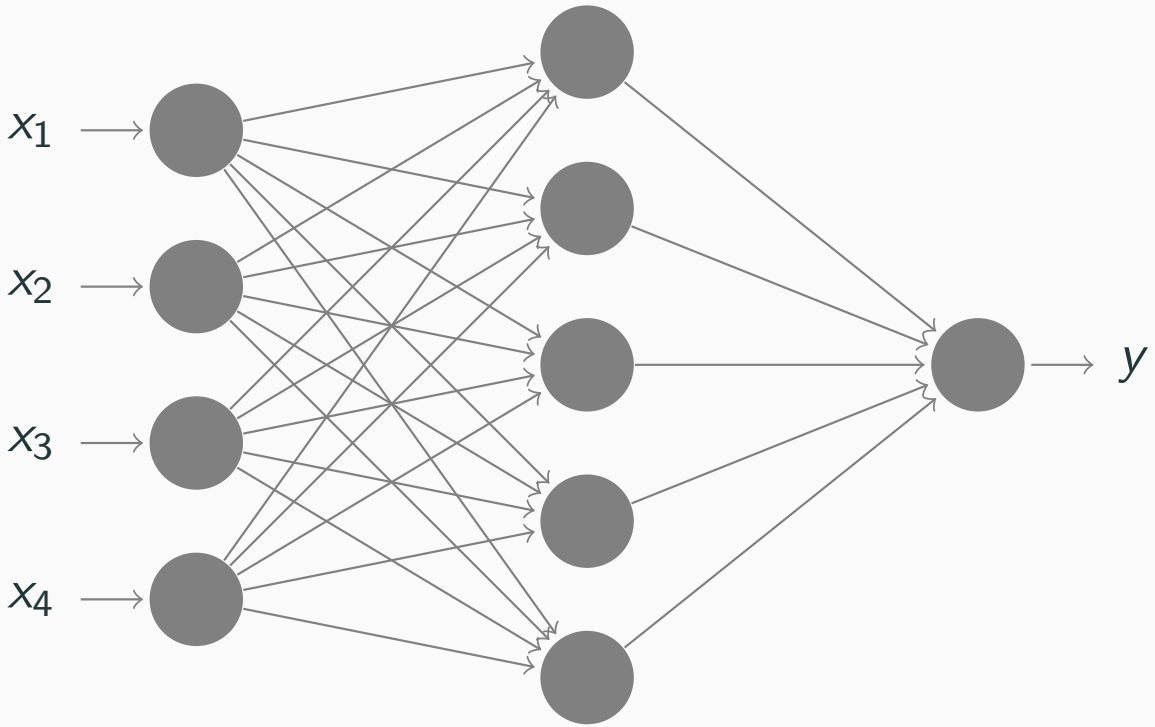
5.2 Experiments with Bayesian Neural Nets (BNNs) I/IV

BNNs are intractable Bayesian regression models with

$$y|\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{y}; F_{\boldsymbol{\theta}}(\boldsymbol{x}), \sigma^2),$$

with $F_{\boldsymbol{\theta}}(\boldsymbol{x})$ defining a non-linear transform of $\boldsymbol{x}$ parameterized by $\boldsymbol{\theta}$. (**Note:** Our experiments use one hidden layer with 50 ReLu neurons.)

$F_{\boldsymbol{\theta}}(\boldsymbol{x})$



5.2 Experiments with Bayesian Neural Nets (BNNs) II/IV

Methods: Comparison of black box approximate Bayesian methods:

- **VI**
- **F-VI** based on $F = D_{AR}^{(\alpha)}$ (Li and Turner, 2016)
- **F-VI** based on $F = D_{A}^{(\alpha)}$ (Hernández-Lobato et al., 2016)
- **GVI** with $D = D_{AR}^{(\alpha)}$.

Note: Everything run with settings of Li and Turner (2016) and Hernández-Lobato et al. (2016)

- Variational family $\mathcal{Q}$: A fully factorized normal
- Optimization of σ^2 (i.e., point estimation akin to type-II ML)
- ADAM (Kingma and Ba, 2014) with default settings and 500 epochs
- 50 Random splits with 90:10 training:test ratio
- benchmark UCI (Lichman et al., 2013) datasets

...

5.2 Experiments with Bayesian Neural Nets (BNNs) III/IV

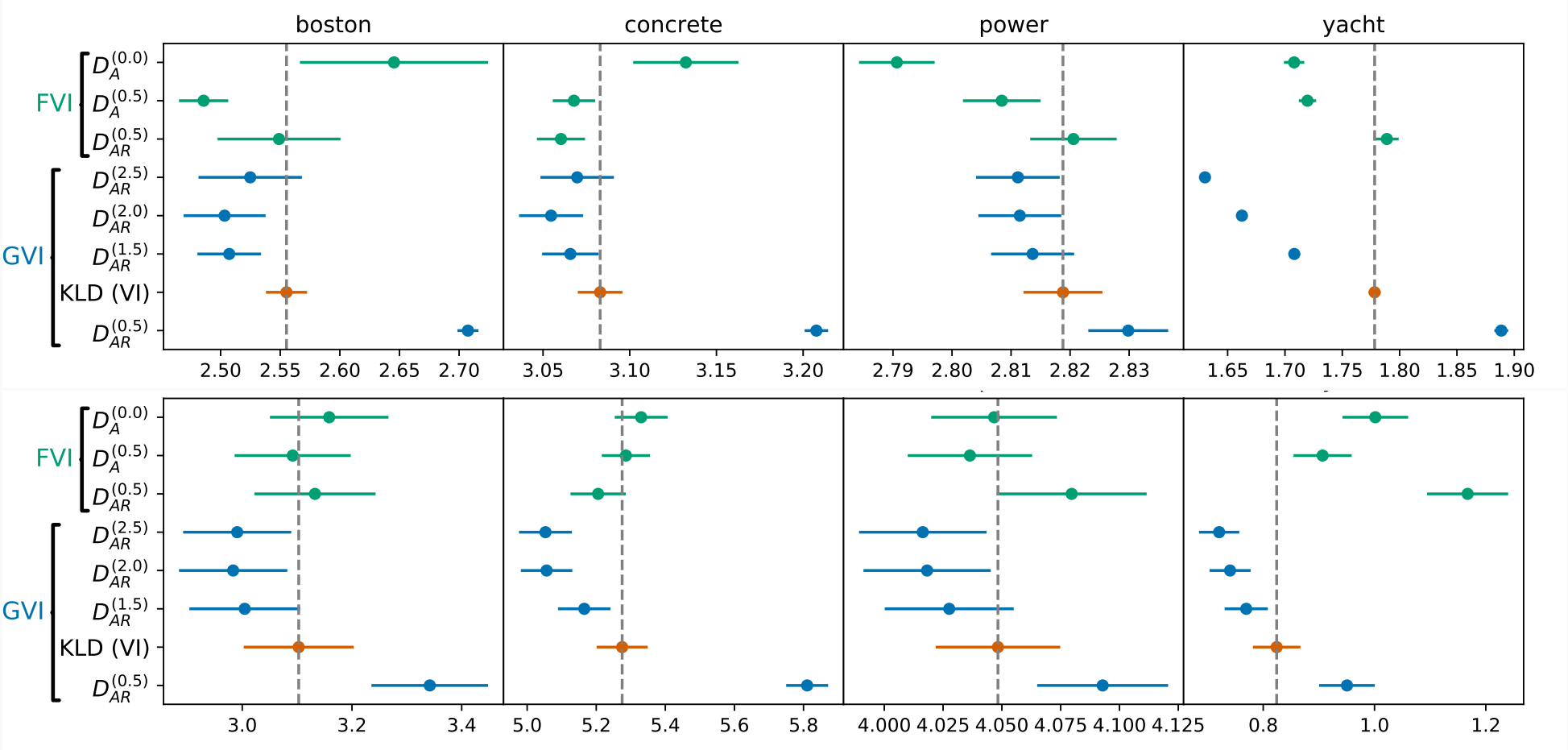


Figure 6 – Performance on BNNs: **F-VI**, **GVI** with $D = D_{AR}^{(\alpha)}$, and **VI**. **Top**: Negative test log likelihoods. **Bottom row**: Test RMSE.

Observation: **GVI** outperforms **VI** for **over-concentrated** posteriors (i.e. $\alpha > 1$)! So how does **under-concentrated** **F-VI** outperform **VI**?!?

5.2 Experiments with Bayesian Neural Nets (BNNs) IV/V

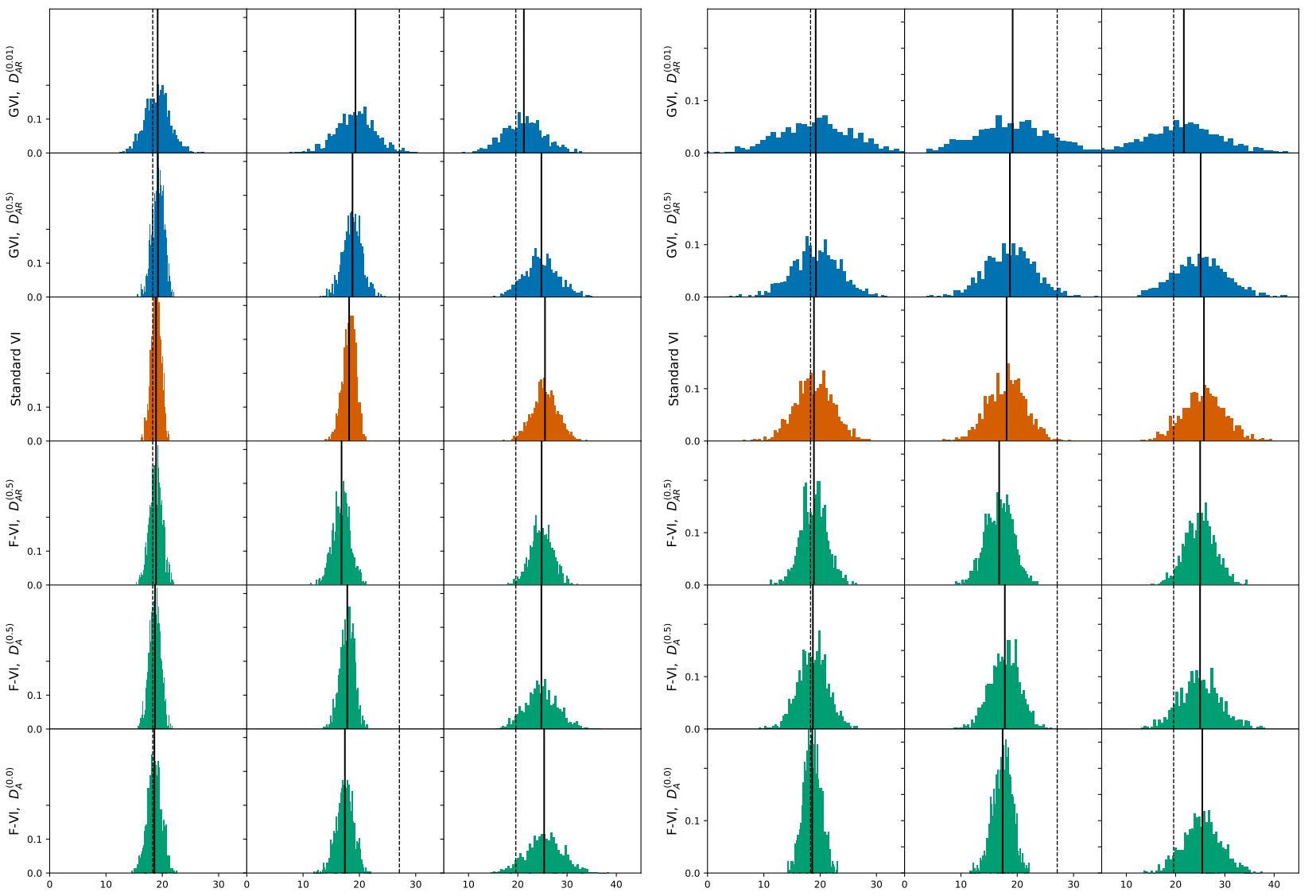


Figure 7 – Left: Parameter posteriors (**F-VI** as expected). **Right:** Posterior predictives (**F-VI not** as expected)

5.2 Experiments with Bayesian Neural Nets (BNNs) V/V

Q: Why does this happen for **F-VI** and not for **GVI**?!

A: **F-VI** does not distinguish uncertainty quantification & loss!

F-VI objective: σ^2 affects **target (!)**

$$\widehat{\sigma}^2, q^*(\boldsymbol{\theta}|\widehat{\sigma}^2, \boldsymbol{\kappa}) = \arg\min_{\sigma^2} \left\{ \arg\min_{q \in \mathcal{Q}} F\left(q(\boldsymbol{\theta}|\sigma^2, \boldsymbol{\kappa}) \| \underbrace{\widetilde{q}(\boldsymbol{\theta}|\sigma^2, \boldsymbol{x}, \boldsymbol{y})}_{\text{i.e., } \widetilde{q}=\widetilde{q}^{\sigma}} \right) \right\}$$

$\Rightarrow$ optimizing for $\sigma^2 =$ **changing the target $\widetilde{q}^{\sigma}$**!

GVI objective: σ^2 indexes the **loss** only

$$\widehat{\sigma}^2, q^*(\boldsymbol{\theta}|\widehat{\sigma}^2, \boldsymbol{x}, \boldsymbol{y}) = \arg\min_{\sigma^2} \left\{ \arg\min_{q \in \mathcal{Q}} \left\{ \mathbb{E}_q \left[\underbrace{\ell_n(\boldsymbol{\theta}, \boldsymbol{x}|\boldsymbol{y}, \sigma^2)}_{\text{i.e., } \ell_n=\ell_n^{\sigma}} \right] + D(q||\pi) \right\} \right\}$$

$\Rightarrow$ optimizing for σ^2 = finding **optimal loss** ℓ_n^{σ}

5.3 Experiments with Deep Gaussian Processes (DGPs) I/II

Principal idea: Use the BNN architecture with GP priors on $F_{\boldsymbol{\theta}}(\cdot)$:

$$
\begin{aligned}
y|\boldsymbol{F}^L &\sim p\left(y\middle|\ \boldsymbol{F}^L\right) \\
\boldsymbol{F}^L|\boldsymbol{F}^{L-1} &\sim \mathrm{GP}\left(\mu^L(\boldsymbol{F}^{L-1}), \mathrm{K}^L(\boldsymbol{F}^{L-1}, \boldsymbol{F}^{L-1})\right) \\
\boldsymbol{F}^{L-1}|\boldsymbol{F}^{L-2} &\sim \mathrm{GP}\left(\mu^{L-1}(\boldsymbol{F}^{L-2}), \mathrm{K}^{L-1}(\boldsymbol{F}^{L-2}, \boldsymbol{F}^{L-2})\right) \\
&\dots \\
\boldsymbol{F}^1|\boldsymbol{x} &\sim \mathrm{GP}\left(\mu^1(\boldsymbol{x}), \mathrm{K}^1(\boldsymbol{x}, \boldsymbol{x})\right),
\end{aligned}
$$

Methods: Comparison of black box approximate Bayesian methods:

- State of the art **VI** (Salimbeni and Deisenroth, 2017) (comprehensively beat competing **F-VI** methods (Bui et al., 2016))
- **GVI** with $\ell_n = \sum_{i=1}^n \mathcal{L}_p^\gamma(\boldsymbol{\theta}, x_i)$.

Note: Everything run with settings of Salimbeni and Deisenroth (2017)

[Derivations for **DGP-GVI**: `https://arxiv.org/abs/1904.02303`.]

5.3 Experiments with Deep Gaussian Processes (DGPs) II/II

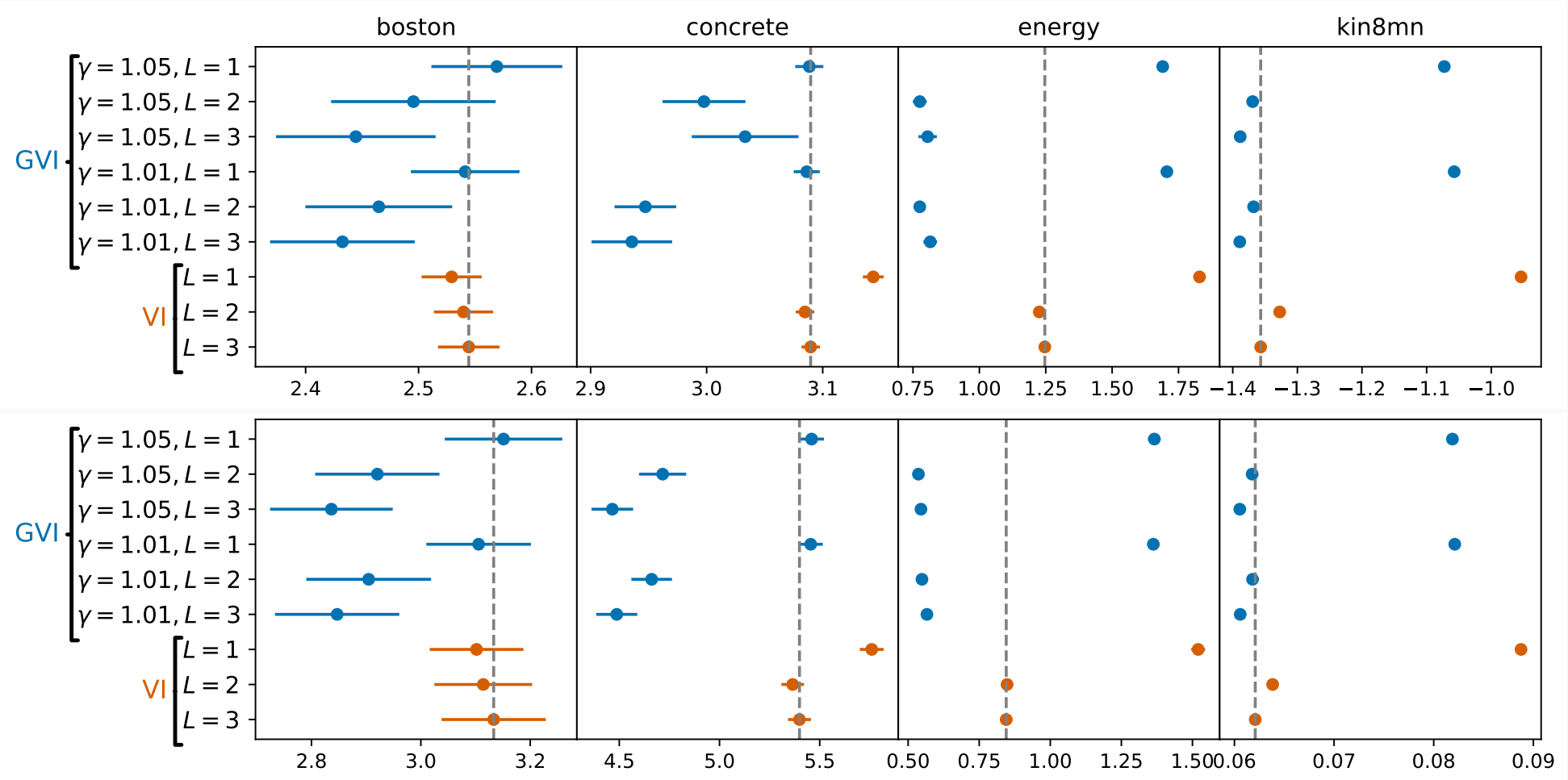


Figure 8 – DGP performance with L layers, **GVI** with $\ell_n(\boldsymbol{\theta}, \boldsymbol{x}) = \sum_{i=1}^{n} \mathcal{L}_p^{\gamma}(\boldsymbol{\theta}, x_i)$ & **VI**. **Top row**: Negative test log likelihoods. **Bottom row**: Test RMSE.

Summary & Conclusion

Summary:

Part 1: Ways to look at Bayesian inference: belief updates (about arbitrary parameters) & optimization over $\mathcal{P}(\boldsymbol{\Theta})$

Part 2: Bayesian inference as a modular & interpretable triplet $P(\ell_n, D, \boldsymbol{\Pi})$: **loss**, **uncertainty quantifier** & **admissible posteriors**.

Part 3: Fallout of $P(\ell_n, D, \boldsymbol{\Pi})$: **VI** optimality & **F-VI** suboptimality $\rightarrow$ **GVI**

Part 4: Some of **GVI**'s use cases: Robust losses, alternative ways of quantifying uncertainty. Also: its upper bound interpretation

Part 5: Black box methods with **GVI** & empirical performance.

Main Conclusions:

(I) **GVI**: **principled & explicit** design of $\mathcal{Q}$-constrained posteriors

(II) **GVI**: **tackles drawbacks of VI** (e.g., robustness, marginals)

(III) **GVI**: **State of the art** $\mathcal{Q}$-constrained posteriors **on BNNs & DGPs**

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Appendix: Choosing robust ℓ_n

$$\mathcal{L}_p^\beta(\boldsymbol{\theta}, x_i) = -\frac{1}{\beta - 1} p(x_i|\boldsymbol{\theta})^{\beta-1} + \frac{l_{p,\beta}(\boldsymbol{\theta})}{\beta}$$

$$\mathcal{L}_p^\gamma(\boldsymbol{\theta}, x_i) = -\frac{1}{\gamma - 1} p(x_i|\boldsymbol{\theta})^{\gamma-1} \frac{\gamma}{l_{p,\gamma}(\boldsymbol{\theta})^{\frac{\gamma-1}{\gamma}}}$$

$$l_{p,c}(\boldsymbol{\theta}) = \int p(x|\boldsymbol{\theta})^c dx$$

where $l_{p,c}(\boldsymbol{\theta}) = \int p(x|\boldsymbol{\theta})^c dx$.

Note 1: $\mathcal{L}_p^\gamma(\boldsymbol{\theta}, x_i)$ multiplicative & always $< 0 \rightarrow$ store as log!

Note 2: Conditional independence $\neq$ additive for $\mathcal{L}_p^\beta(\boldsymbol{\theta}, x_i), \mathcal{L}_p^\gamma(\boldsymbol{\theta}, x_i)$

Note 3: In practice, usually best to choose $\beta/\gamma = 1 + \varepsilon$ for some small ε

Appendix: Choosing hyperparameters

Q: Any principled way of choosing hyperparameters?

A: Very much unsolved problem, solutions so far:

- D: brute force (CV) (Regli and Silva, 2018) [slow/expensive]
- ℓ_n: Via *points of highest influence* (Knoblauch et al., 2018)
- ℓ_n: on-line updates using loss-minimization (Knoblauch et al., 2018)

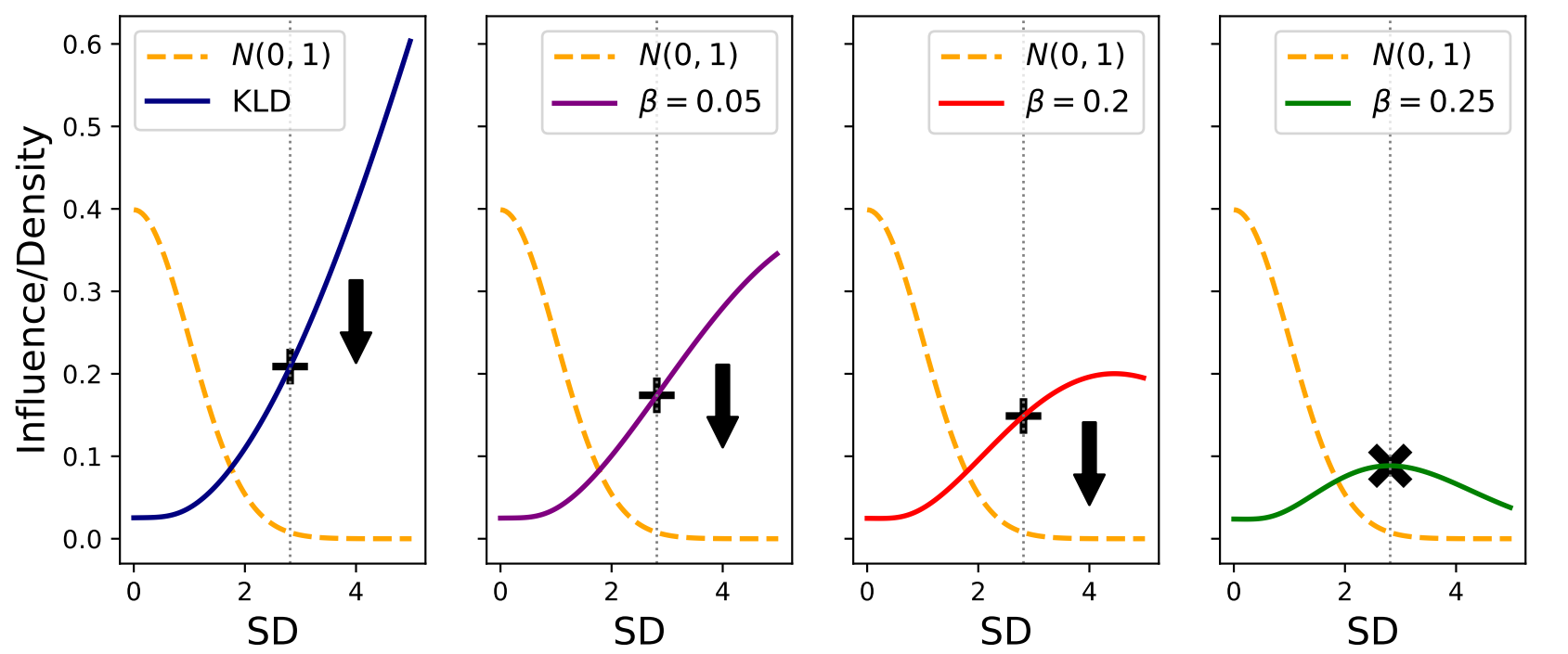


Figure 9 – Illustration of the initialization procedure using *points of highest influence* logic, from left to right.

Appendix: Choosing D for conservative marginals I/II

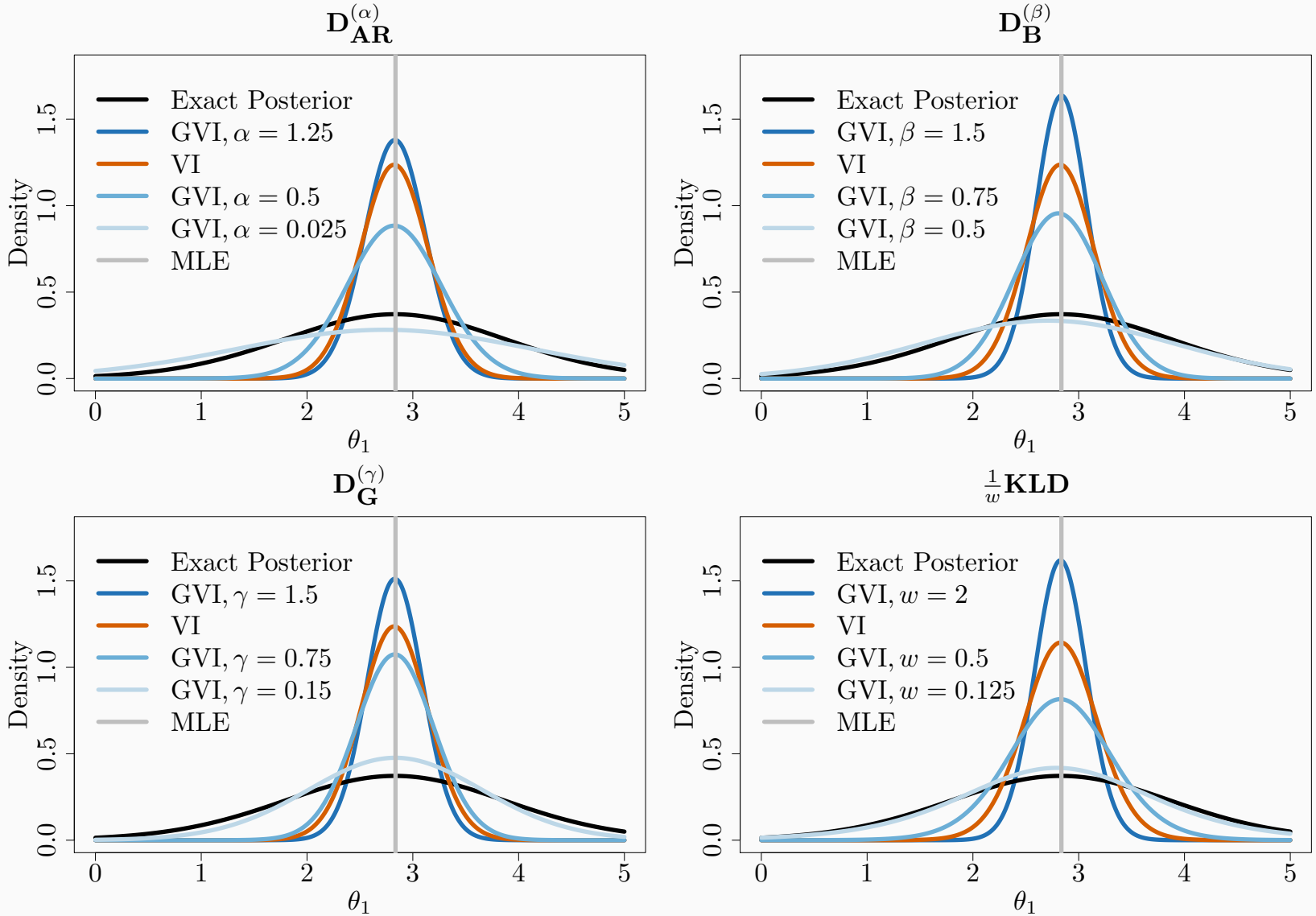


Figure 10 – Marginal VI and GVI posterior for a Bayesian linear model under the $D_{AR}^{(\alpha)}$, $D_{B}^{(\beta)}$, $D_{G}^{(\gamma)}$ and $\frac{1}{w}\mathrm{KLD}$ uncertainty quantifier for different values of the divergence hyperparameters.

Appendix: Choosing D for conservative marginals II/II

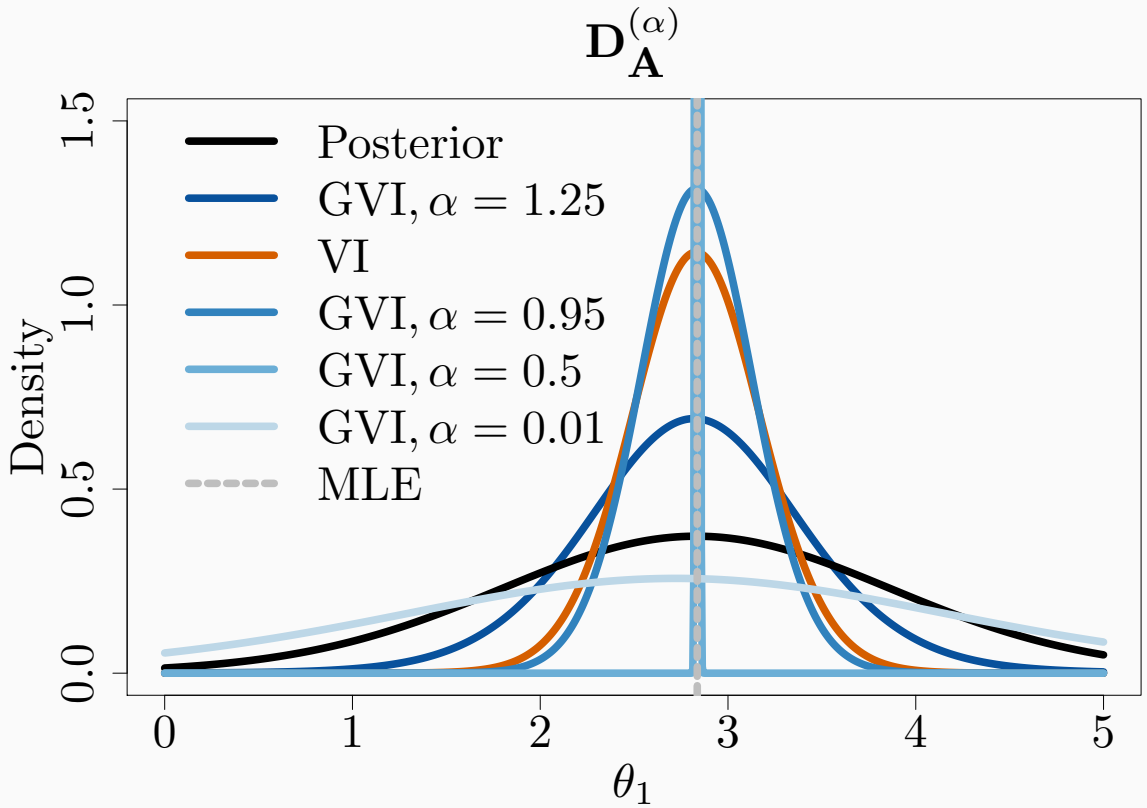


Figure 11 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under the $D_A^{(\alpha)}$ uncertainty quantifier. The boundedness of the $D_A^{(\alpha)}$ causes **GVI** to severely over-concentrate if α is not carefully specified.

Appendix: Choosing D for prior robustness I/IV

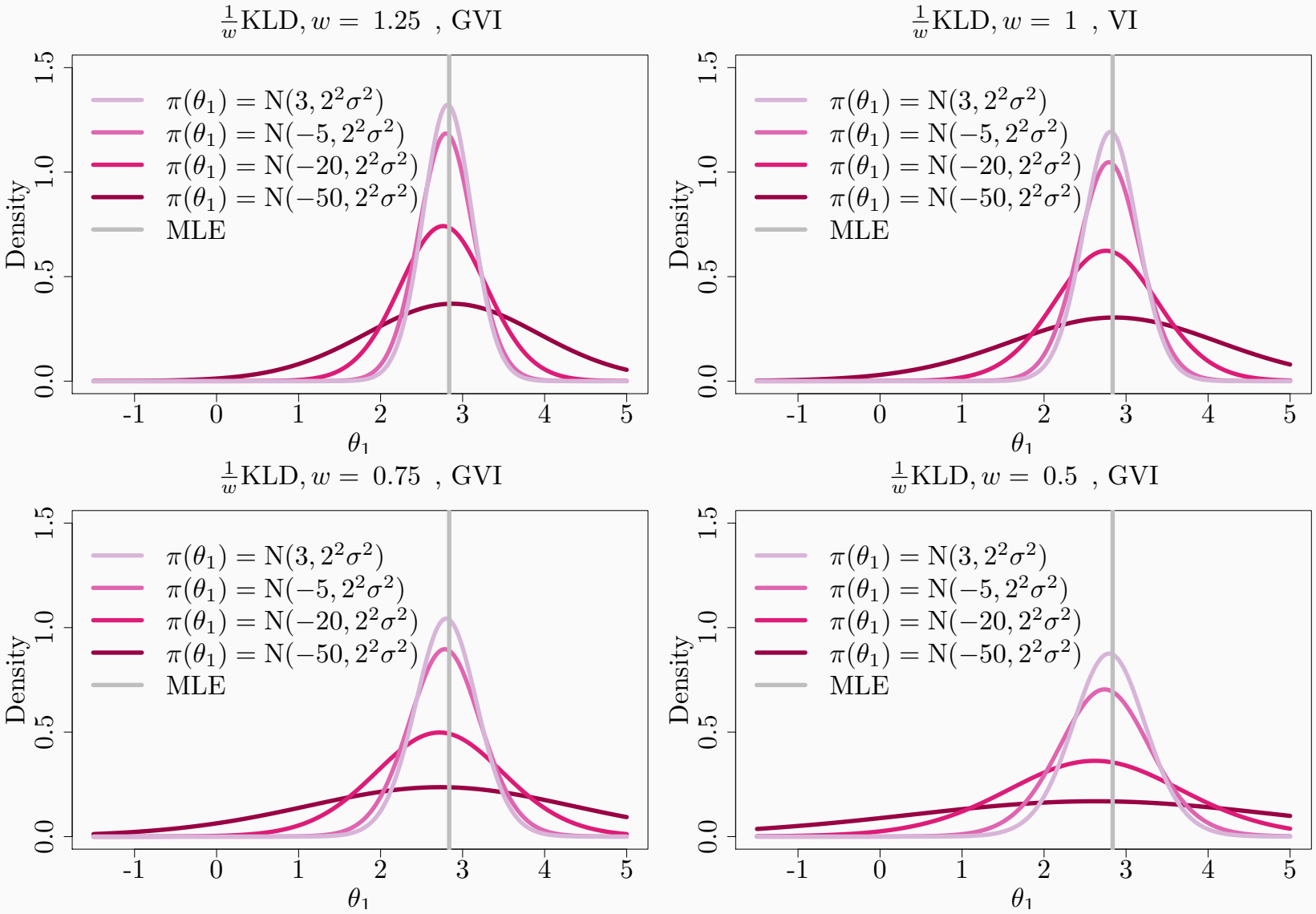


Figure 12 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under different priors, using $D = \frac{1}{w}\mathrm{KLD}$ as the uncertainty quantifier.

Appendix: Choosing D for prior robustness II/IV

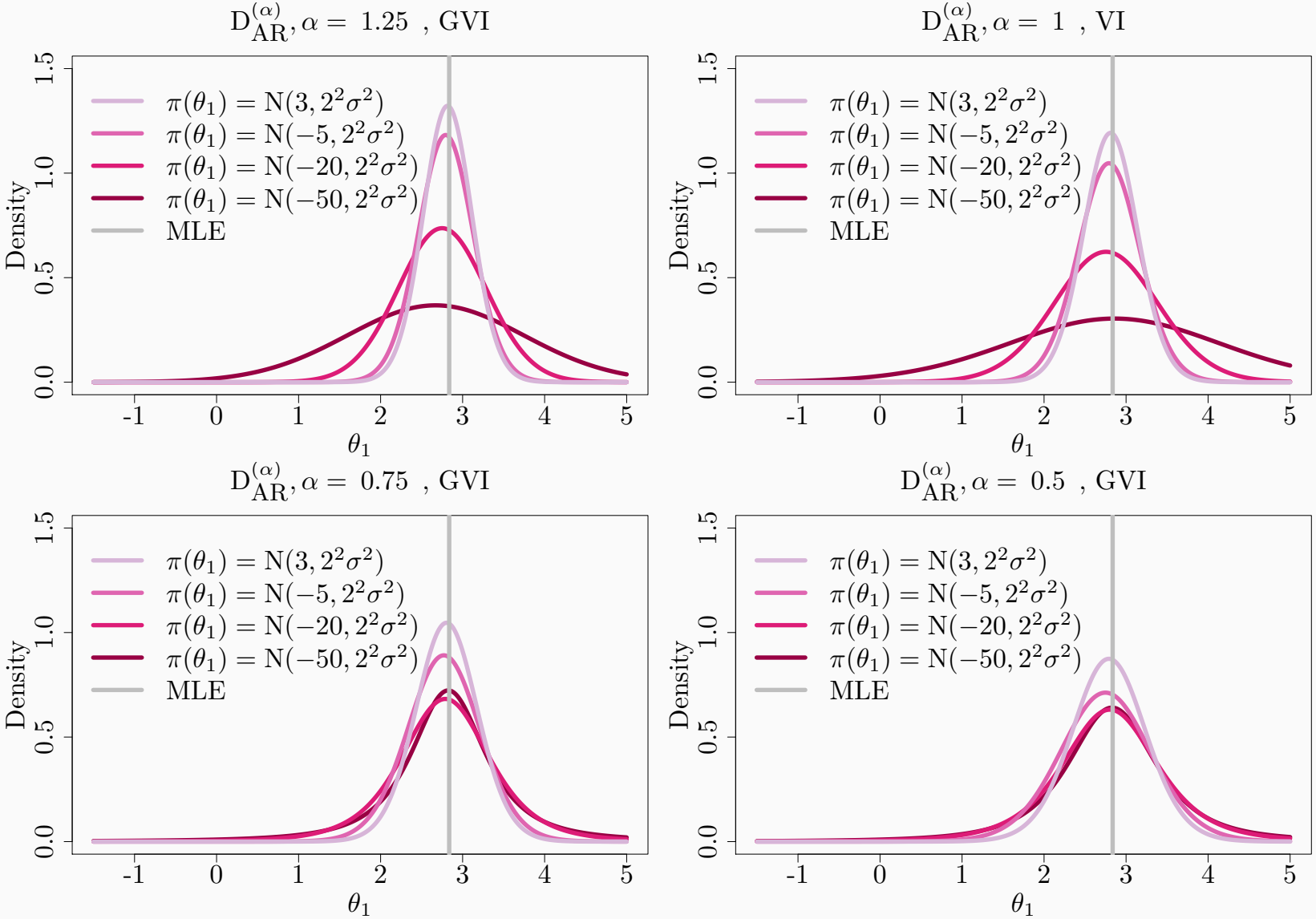


Figure 13 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under different priors, using $D = D_{AR}^{(\alpha)}$ as the uncertainty quantifier.

Appendix: Choosing D for prior robustness III/IV

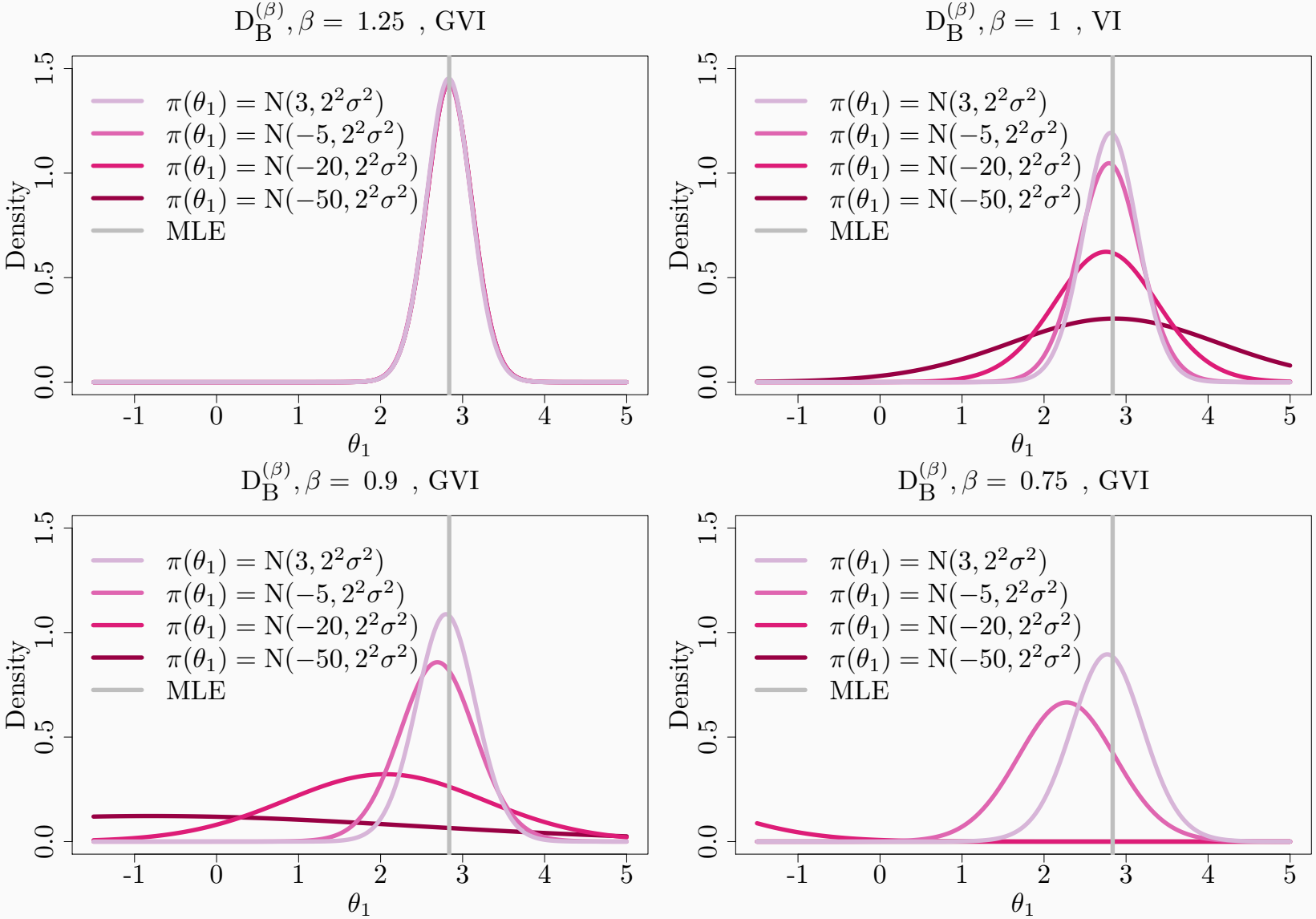


Figure 14 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under different priors, using $D = D_B^{(\beta)}$ as the uncertainty quantifier.

Appendix: Choosing D for prior robustness IV/IV

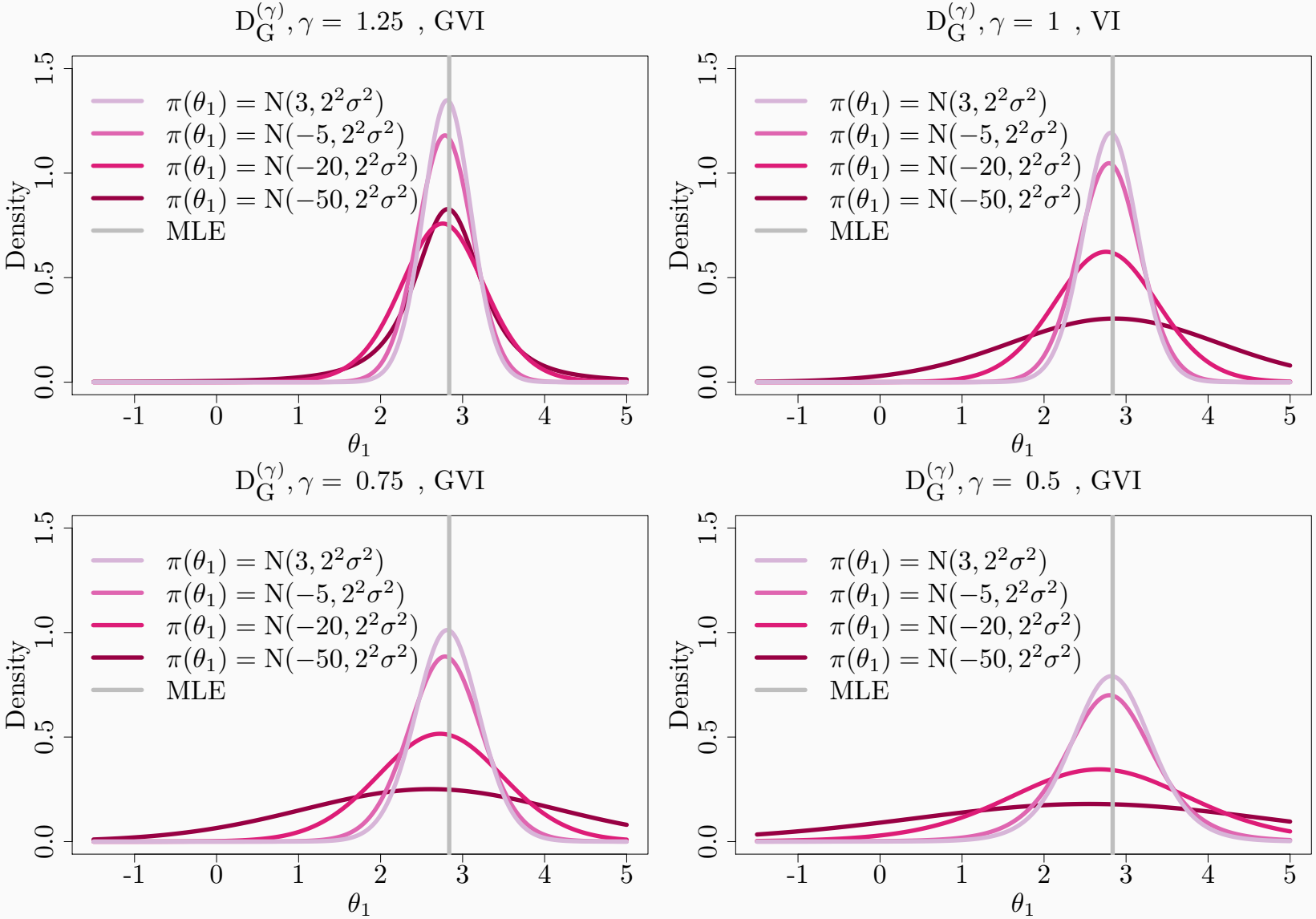


Figure 15 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under different priors, using $D = D_G^{(\gamma)}$ as the uncertainty quantifier.

Appendix: GVI lower bound interpretation I/II

Question: VI is also interpretable as optimizing a lower bound on the evidence! Is there anything comparable for GVI?
Answer: Yes, e.g. for $D_B^{(\beta)}$, $D_G^{(\gamma)}$, $D_{AR}^{(\alpha)}$: Consider *generalized* evidence:

Recall: Generalized Bayes posterior (Bissiri et al., 2016) is

$$q^*_{\ell_n}(\boldsymbol{\theta}) \propto \pi(\boldsymbol{\theta}) \exp\{-\ell_n(\boldsymbol{\theta}, \boldsymbol{x})\} \quad \text{and so } p_{\ell_n}(\boldsymbol{x}) = \int_{\boldsymbol{\Theta}} q^*_{\ell_n}(\boldsymbol{\theta}) d\boldsymbol{\theta}$$

GVI's objectives $L(q|\boldsymbol{x}, D, \ell_n)$ will optimize

$$L(q|\boldsymbol{x}, D, \ell_n) \geq g^D(\underbrace{-\log p_{f^D(\ell_n)}(\boldsymbol{x})}_{\substack{\text{negative log evidence;} \\ f^D(\ell_n) \text{ maps } \ell_n \text{ into a new loss}}}) + \underbrace{T^D(q)}_{\text{Approximate target}}$$

(**Note**: VI is special case where this holds with *equality* (so that the approximate target is the exact target) and where $g^{\mathrm{KLD}}(x) = x$, $L(q|\boldsymbol{x}, D, \ell_n) = \mathrm{ELBO}(q)$, $T^{\mathrm{KLD}}(q) = \mathrm{KLD}(q||q^*_{\ell_n})$, $f^{\mathrm{KLD}}(\ell_n) = \ell_n$.)

Appendix: GVI lower bound interpretation II/II

GVI's objectives $L(q|\boldsymbol{x}, D, \ell_n)$ will optimize

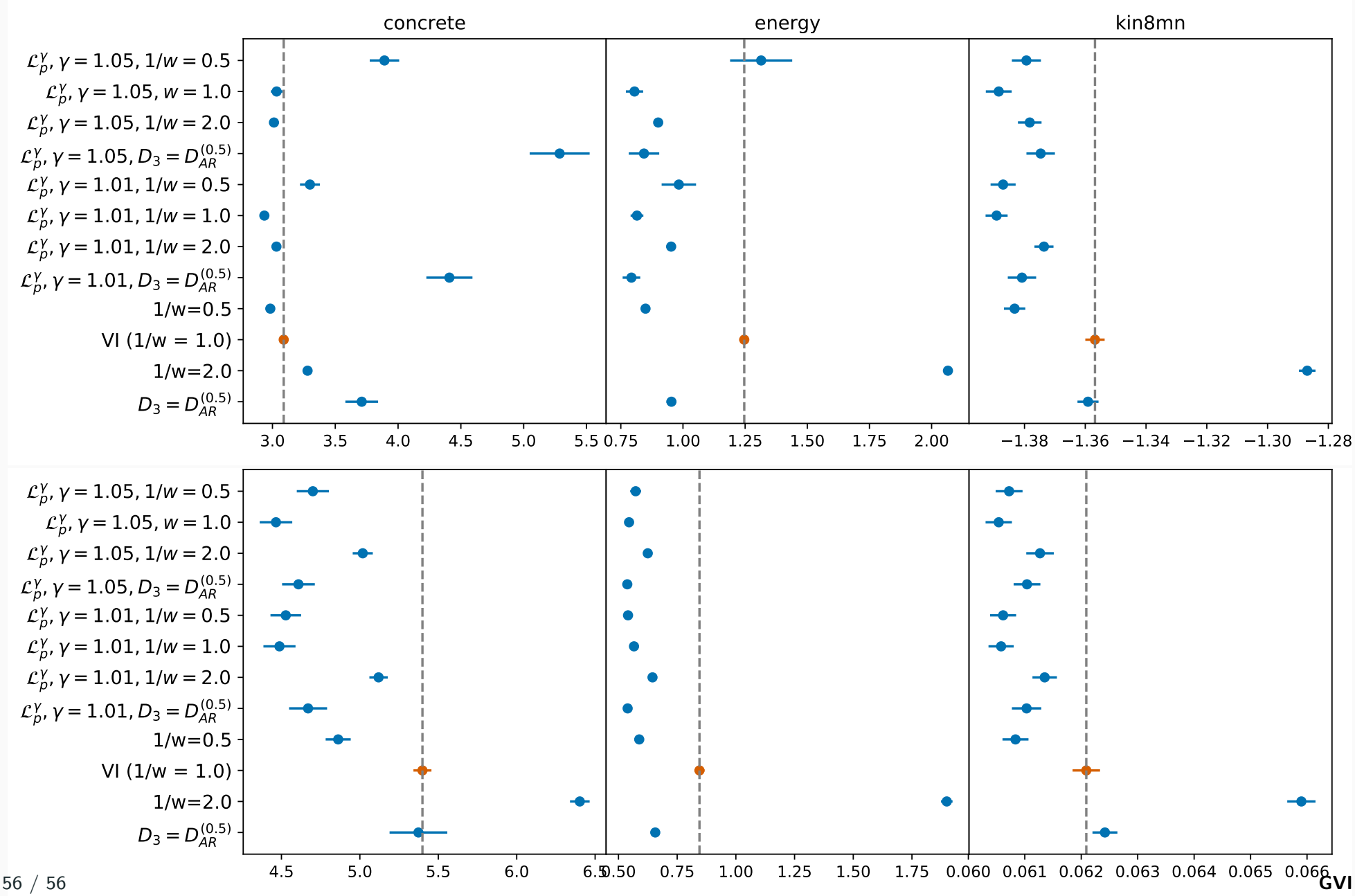
$$L(q|\boldsymbol{x}, D, \ell_n) \geq g^D(\underbrace{-\log p_{f^D(\ell_n)}(\boldsymbol{x})}_{\substack{\text{negative log evidence;} \\ f^D(\ell_n) \text{ maps } \ell_n \text{ into a new loss}}}) + \underbrace{T^D(q)}_{\text{Approximate target}}$$

Example: Rényi's α-divergence ($D_{AR}^{(\alpha)}$) for $\alpha > 1$ gives

$$g^{D_{AR}^{(\alpha)}}(x) = \frac{1}{\alpha}x,$$
$$f^{D_{AR}^{(\alpha)}}(\ell_n) = \alpha\ell_n,$$
$$T_{D_{AR}^{(\alpha)}}(q) = \frac{1}{\alpha}\mathrm{KLD}(q||q^*_{\alpha\ell_n}),$$

so putting it together one finds that for $D = D_{AR}^{(\alpha)}$ with $\alpha > 1$,

$$L(q|\boldsymbol{x}, D, \ell_n) \geq -\frac{1}{\alpha}\log p_{\alpha\ell_n}(\boldsymbol{x}) + \frac{1}{\alpha}\mathrm{KLD}(q||q^*_{\alpha\ell_n})$$

(Which is just a $\frac{1}{\alpha}$-scaled version of the ELBO for the loss $\alpha\ell_n$!)

5.3 Experiments with Deep Gaussian Processes (DGPs) III/IV